

```
In [1]: # pip install --upgrade seaborn
```

```
In [2]: import warnings
warnings.filterwarnings("ignore", category=FutureWarning)
```

```
In [4]: import seaborn as sns
sns.get_dataset_names()
```

```
Out[4]: ['anagrams',
        'anscombe',
        'attention',
        'brain_networks',
        'car_crashes',
        'diamonds',
        'dots',
        'dowjones',
        'exercise',
        'flights',
        'fmri',
        'geyser',
        'glue',
        'healthexp',
        'iris',
        'mpg',
        'penguins',
        'planets',
        'seaice',
        'taxi',
        'tips',
        'titanic']
```

```
In [5]: import seaborn as sns
import matplotlib.pyplot as plt

# Load sample dataset
tips = sns.load_dataset("tips")

# Set a visual theme
sns.set_theme(style="darkgrid")
```

```
In [6]: tips
```

Out[6]:

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

244 rows × 7 columns

```
In [7]: # Save to CSV
tips.to_csv("tips_dataset.csv", index=False)
```

```
In [8]: import os
os.getcwd()
```

Out[8]: 'C:\\Users\\3star\\AK'

```
In [9]: plt.figure(figsize=(8, 6))
```

Out[9]: <Figure size 800x600 with 0 Axes>
<Figure size 800x600 with 0 Axes>

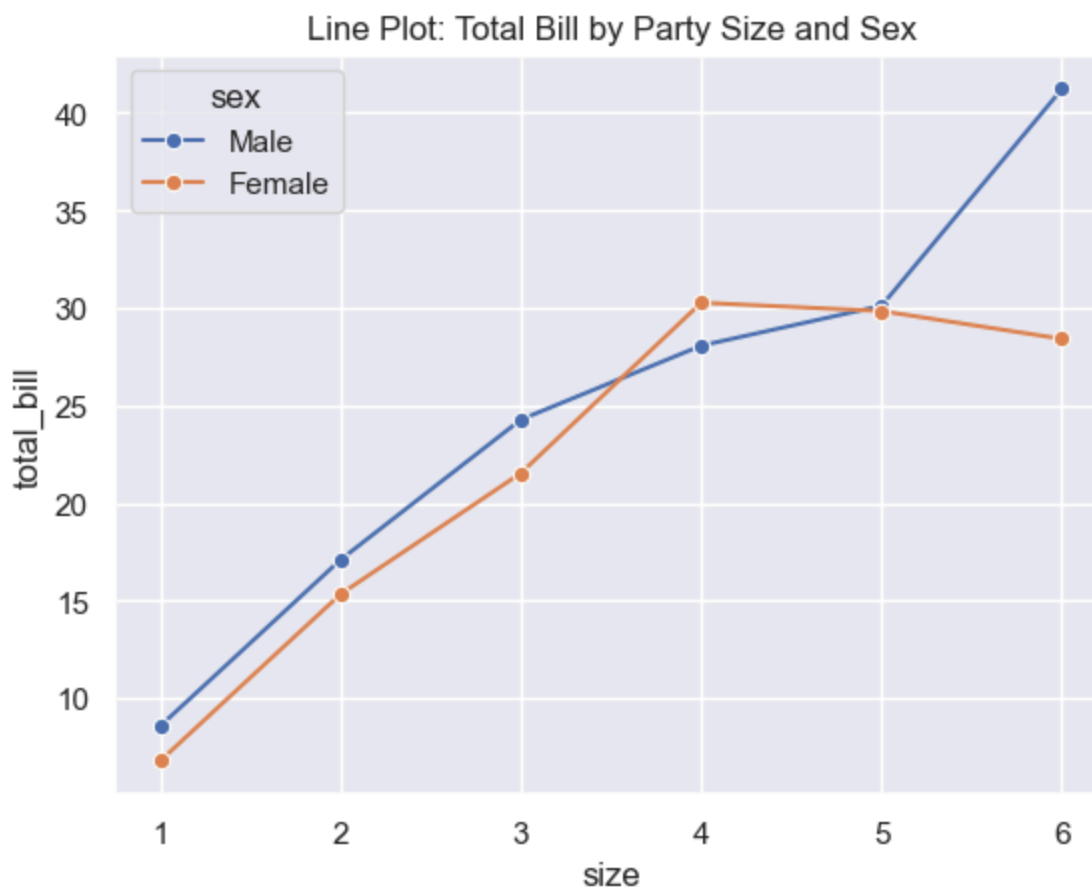
```
In [10]: #1. Scatter Plot: Total Bill vs Tip, with hue and

sns.scatterplot(data=tips, x="total_bill", y="tip", hue="time", size="size", palette=
plt.title("Scatter Plot: Total Bill vs Tip by Time and Size")
plt.show()
```



In [11]: # 2. Line Plot: Total Bill by Party Size, split by Sex

```
sns.lineplot(data=tips, x="size", y="total_bill", hue="sex", ci=None, marker="o")
plt.title("Line Plot: Total Bill by Party Size and Sex")
plt.show()
```

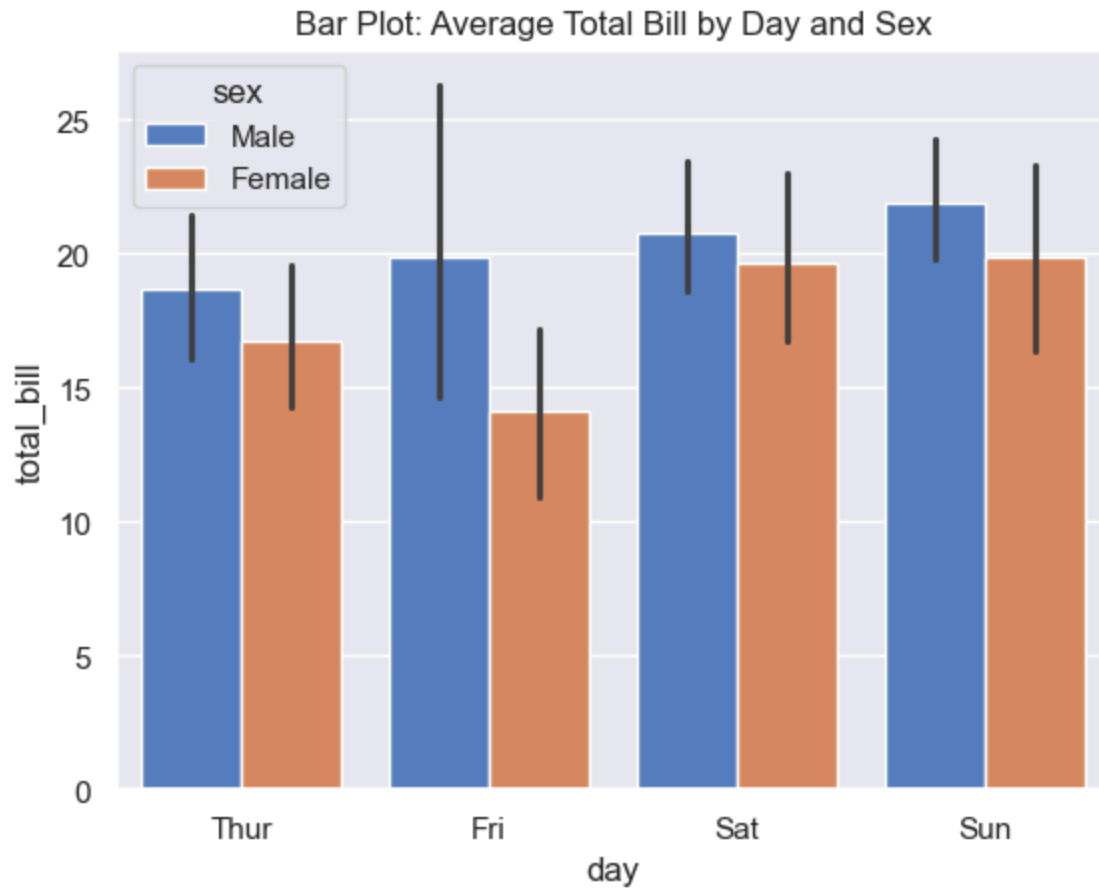


```
In [12]: tips.columns
```

```
Out[12]: Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='object')
```

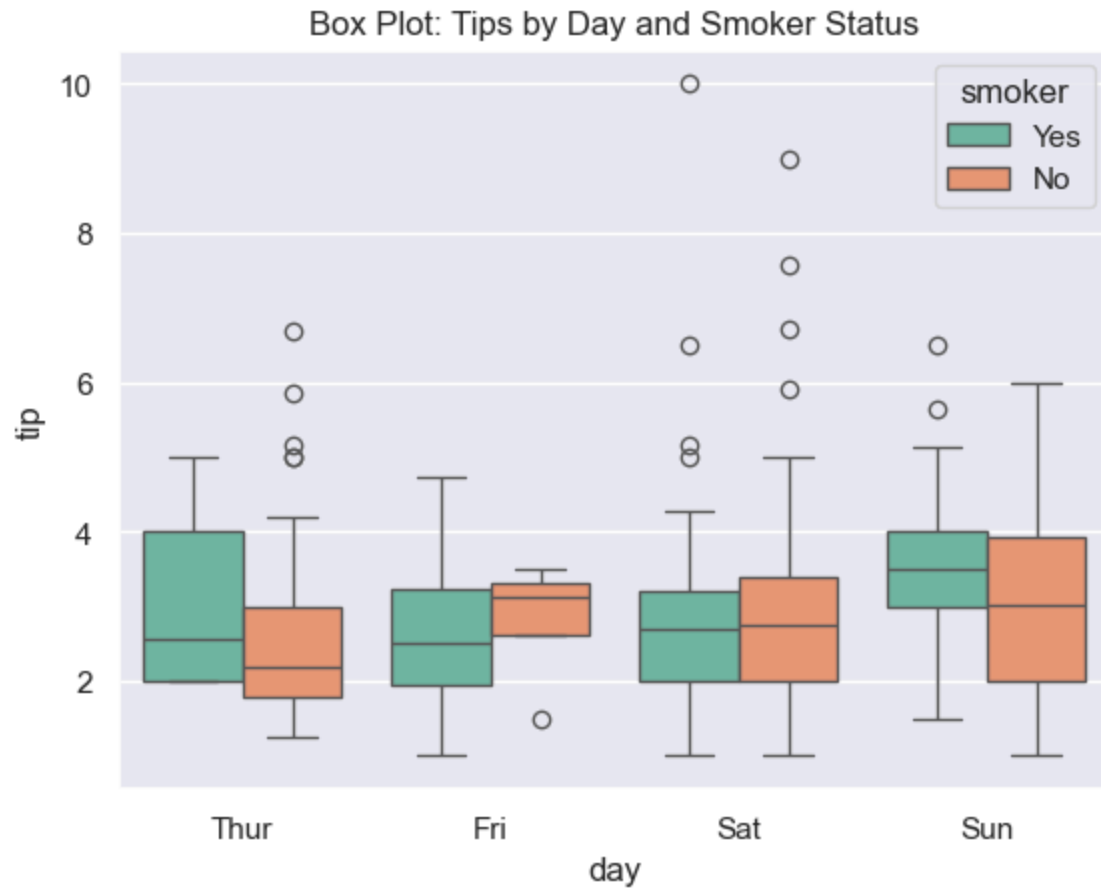
```
In [13]: # 3. Bar Plot: Average Total Bill by Day, split by Sex
```

```
sns.barplot(data=tips, x="day", y="total_bill", hue="sex", palette="muted")  
plt.title("Bar Plot: Average Total Bill by Day and Sex")  
plt.show()
```



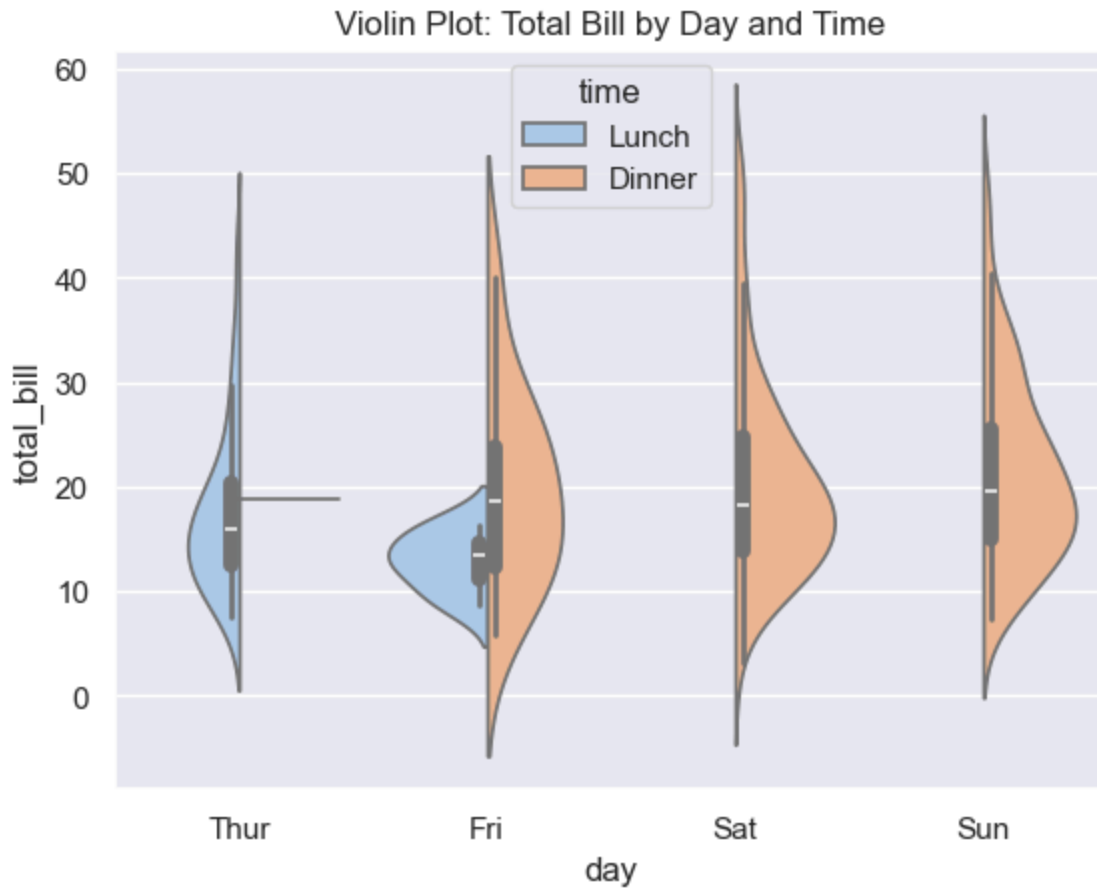
```
In [14]: # 4. Box Plot: Total Bill by Day, split by Time

sns.boxplot(data=tips, x="day", y="tip", hue="smoker", palette="Set2")
plt.title("Box Plot: Tips by Day and Smoker Status")
plt.show()
```



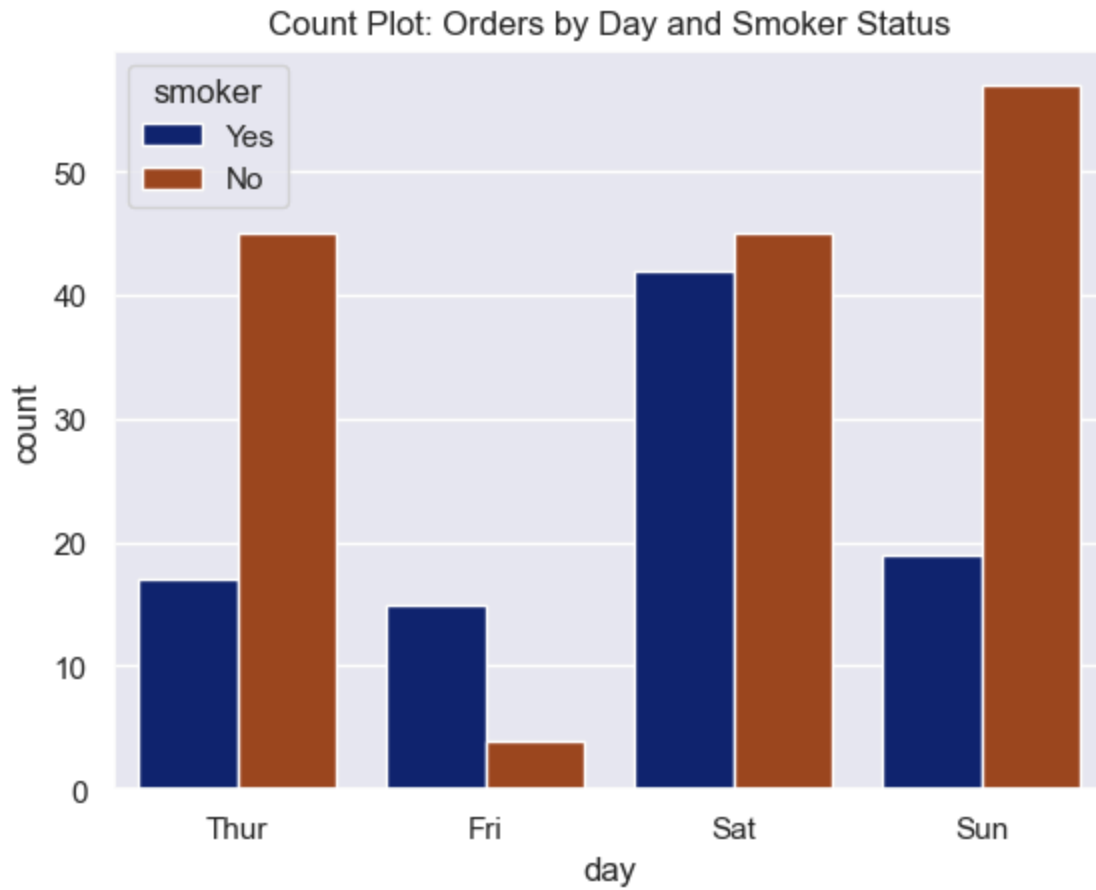
```
In [15]: # 5. Violin Plot: Total Bill by Day, split by Time

sns.violinplot(data=tips, x="day", y="total_bill", hue="time", split=True, palette=
plt.title("Violin Plot: Total Bill by Day and Time")
plt.show()
```



```
In [16]: # 6. Count Plot: Orders by Day, split by Smoker

sns.countplot(data=tips, x="day", hue="smoker", palette="dark")
plt.title("Count Plot: Orders by Day and Smoker Status")
plt.show()
```



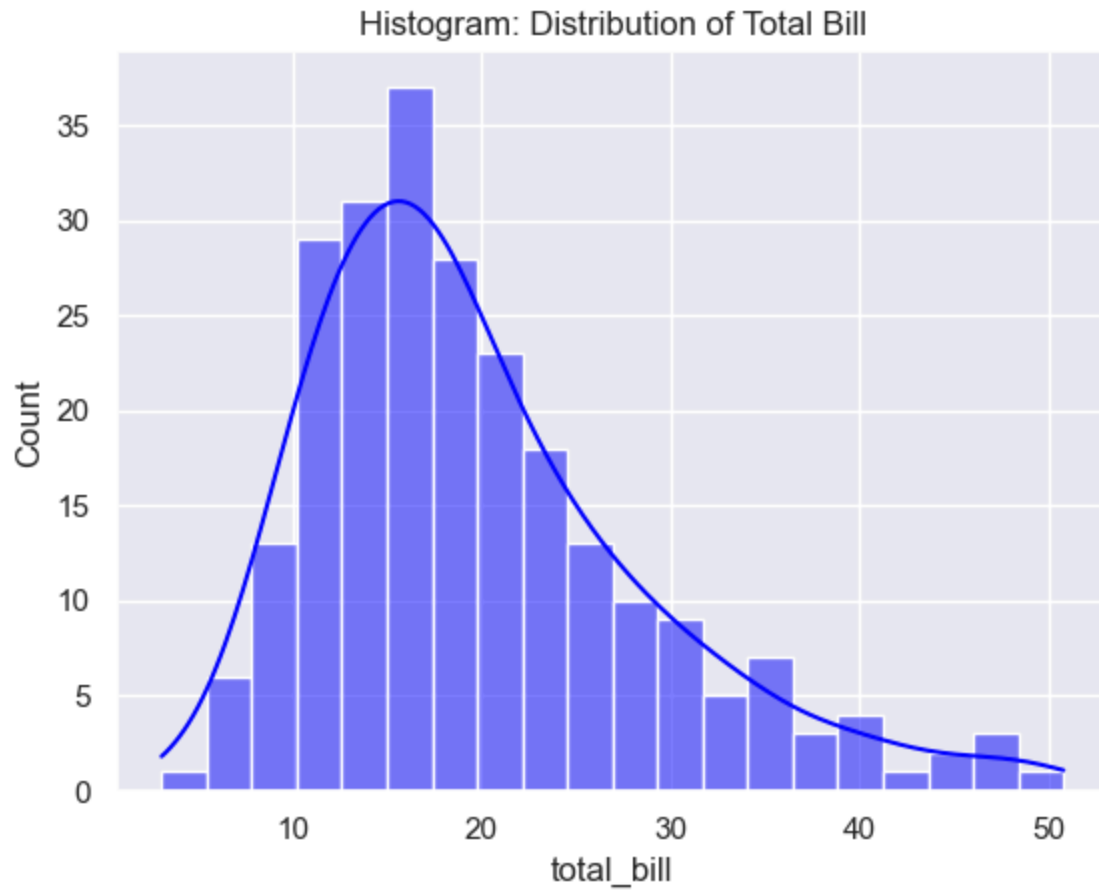
```
In [17]: # 7. Regression Plot: Total Bill vs Tip with regression line

sns.regplot(data=tips, x="total_bill", y="tip", scatter_kws={"s": 50}, line_kws={"c": "red", "dash": [5, 5]})
plt.title("Regression Plot: Total Bill vs Tip")
plt.show()
```



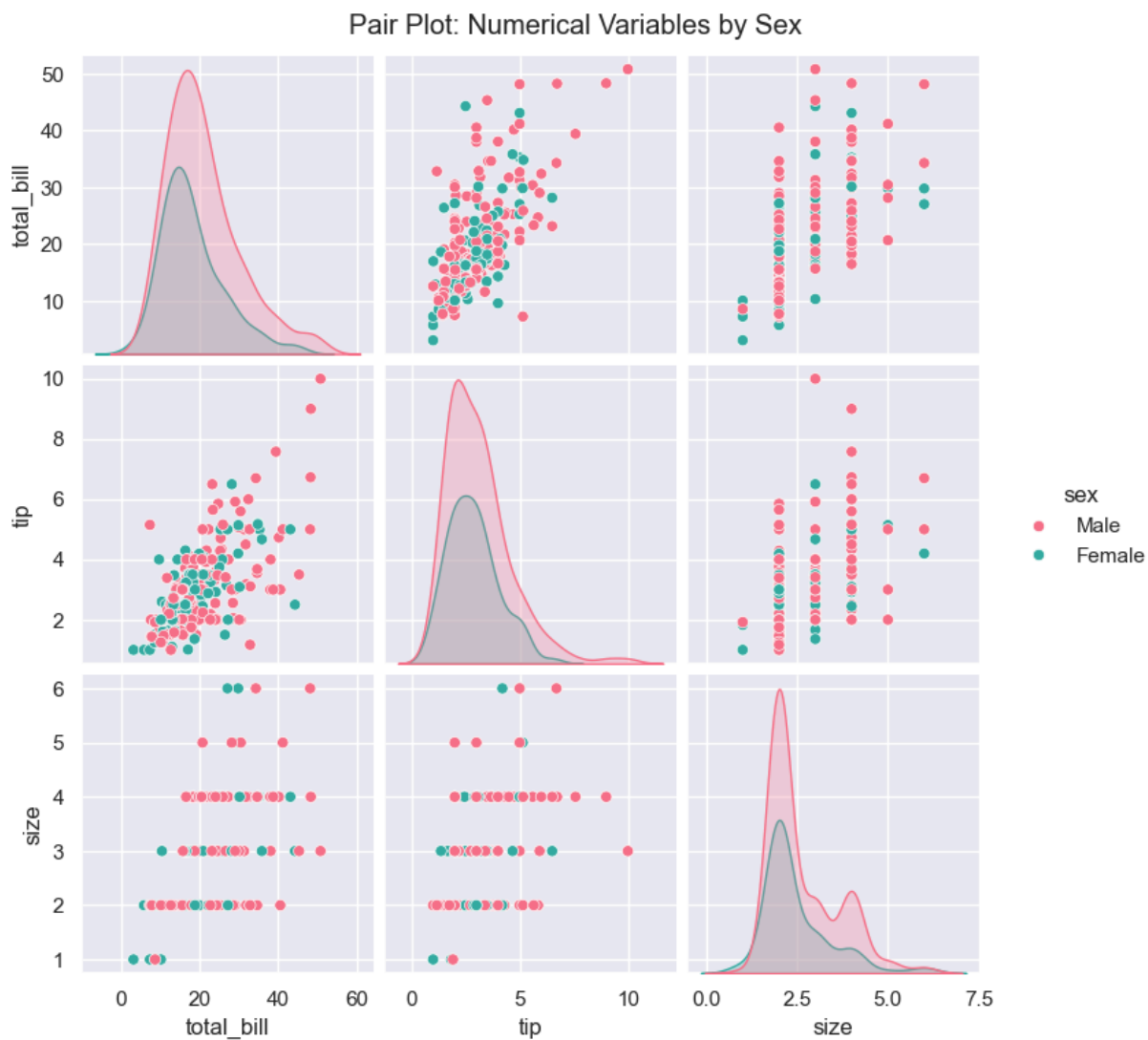
In [18]: #8. Histogram (Distribution Plot): Total Bill with KDE

```
sns.histplot(data=tips, x="total_bill", kde=True, bins=20, color="blue")  
plt.title("Histogram: Distribution of Total Bill")  
plt.show()
```



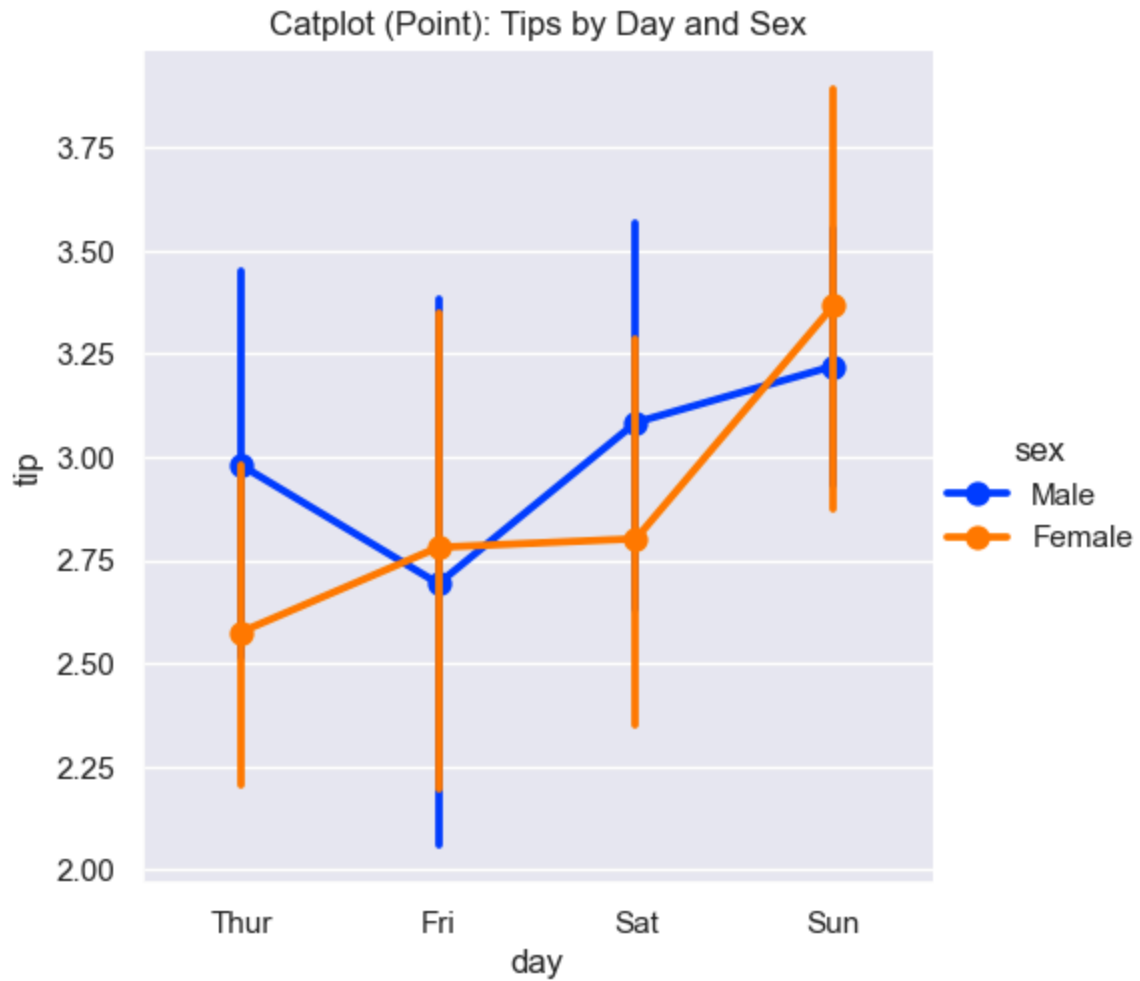
In [19]: *#9. Pair Plot: Relationships between numerical variables*

```
sns.pairplot(tips, hue="sex", vars=["total_bill", "tip", "size"], palette="husl")
plt.suptitle("Pair Plot: Numerical Variables by Sex", y=1.02)
plt.show()
```



In [20]: # 10. Catplot (Point Plot): Tip by Day and Sex

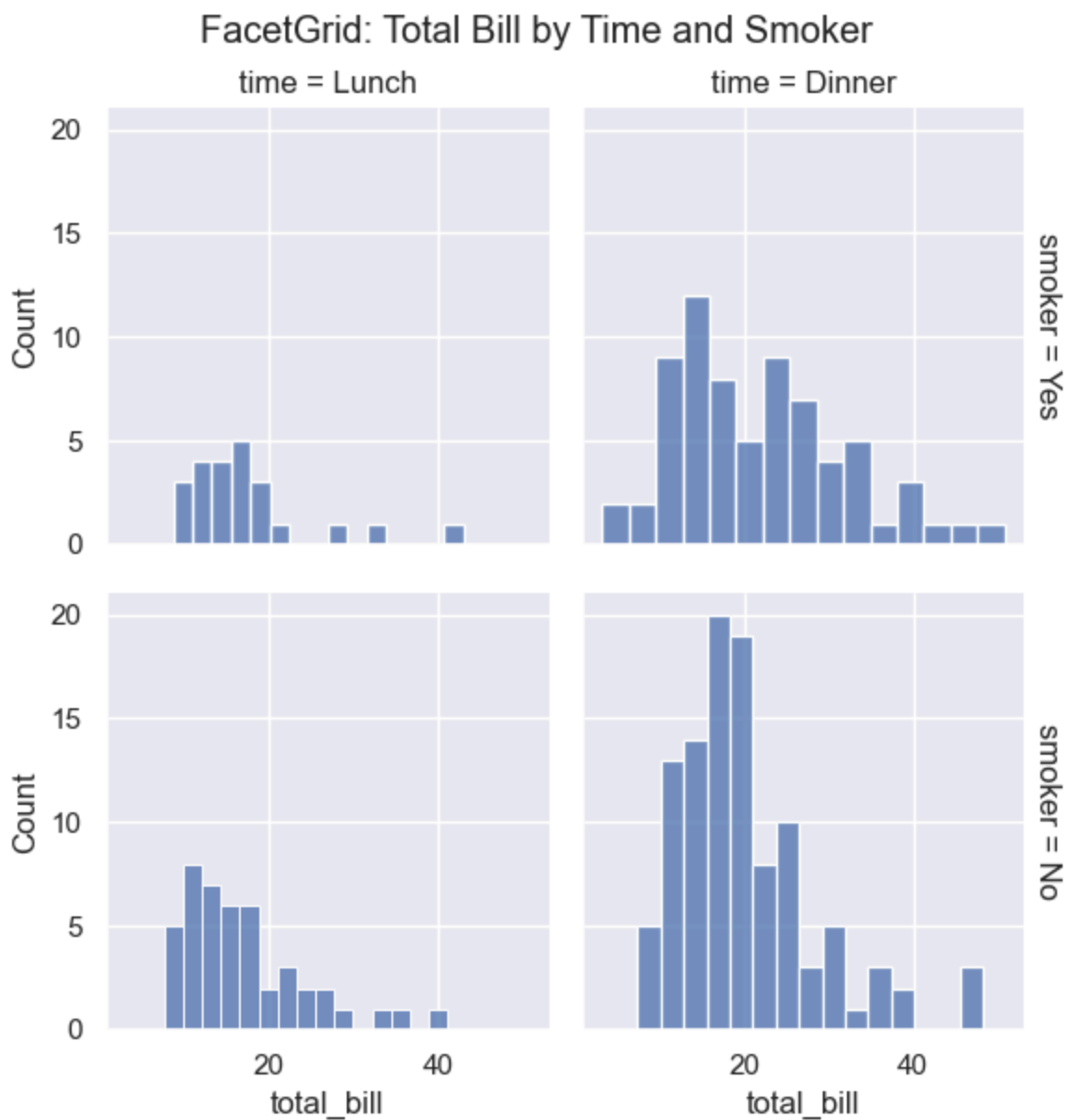
```
sns.catplot(data=tips, x="day", y="tip", hue="sex", kind="point", palette="bright")
plt.title("Catplot (Point): Tips by Day and Sex")
plt.show()
```



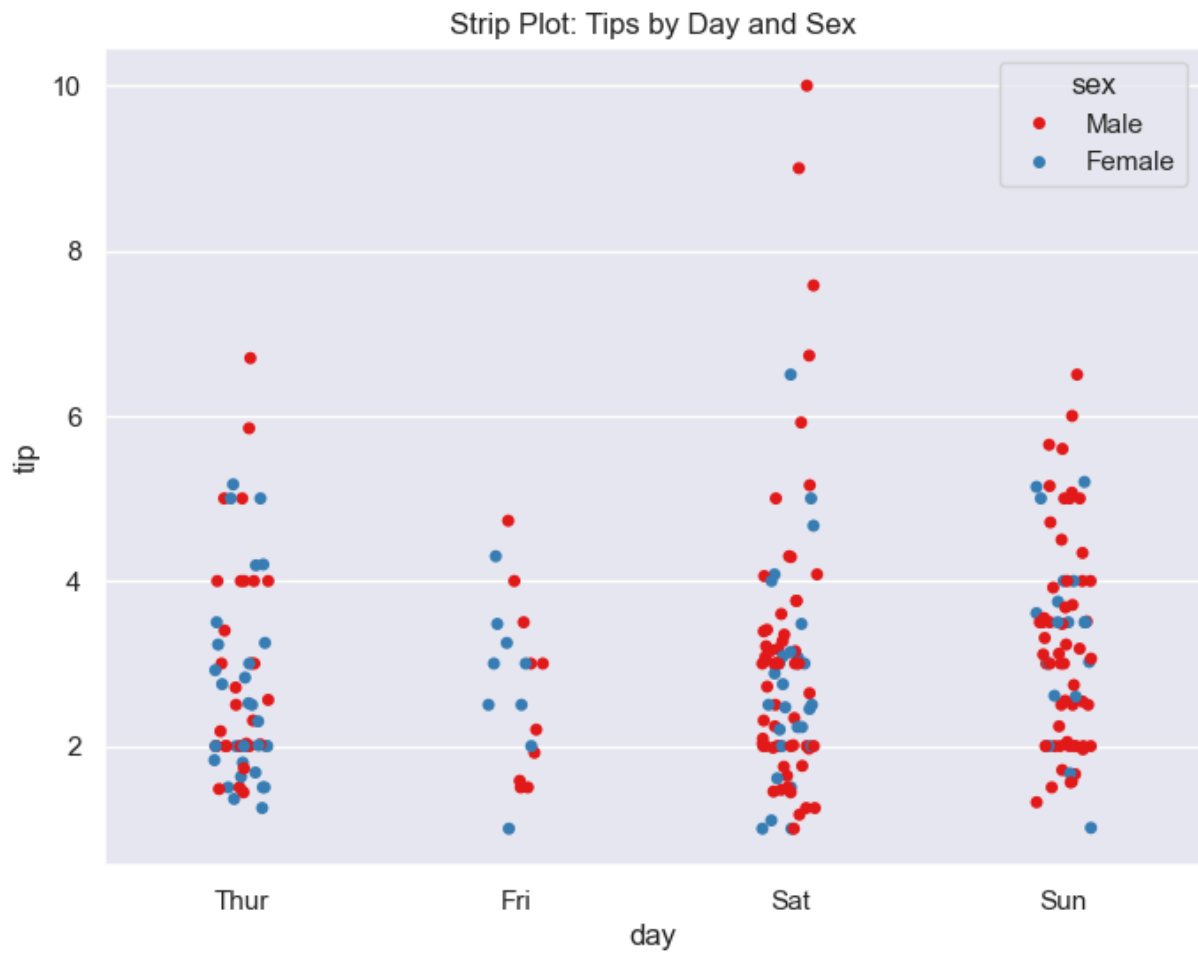
```
In [21]: # 11. Joint Plot: Total Bill vs Tip with marginal distributions
sns.jointplot(data=tips, x="total_bill", y="tip", kind="scatter", hue="smoker", palette="magma")
plt.suptitle("Joint Plot: Total Bill vs Tip by Smoker", y=1.02)
plt.show()
```



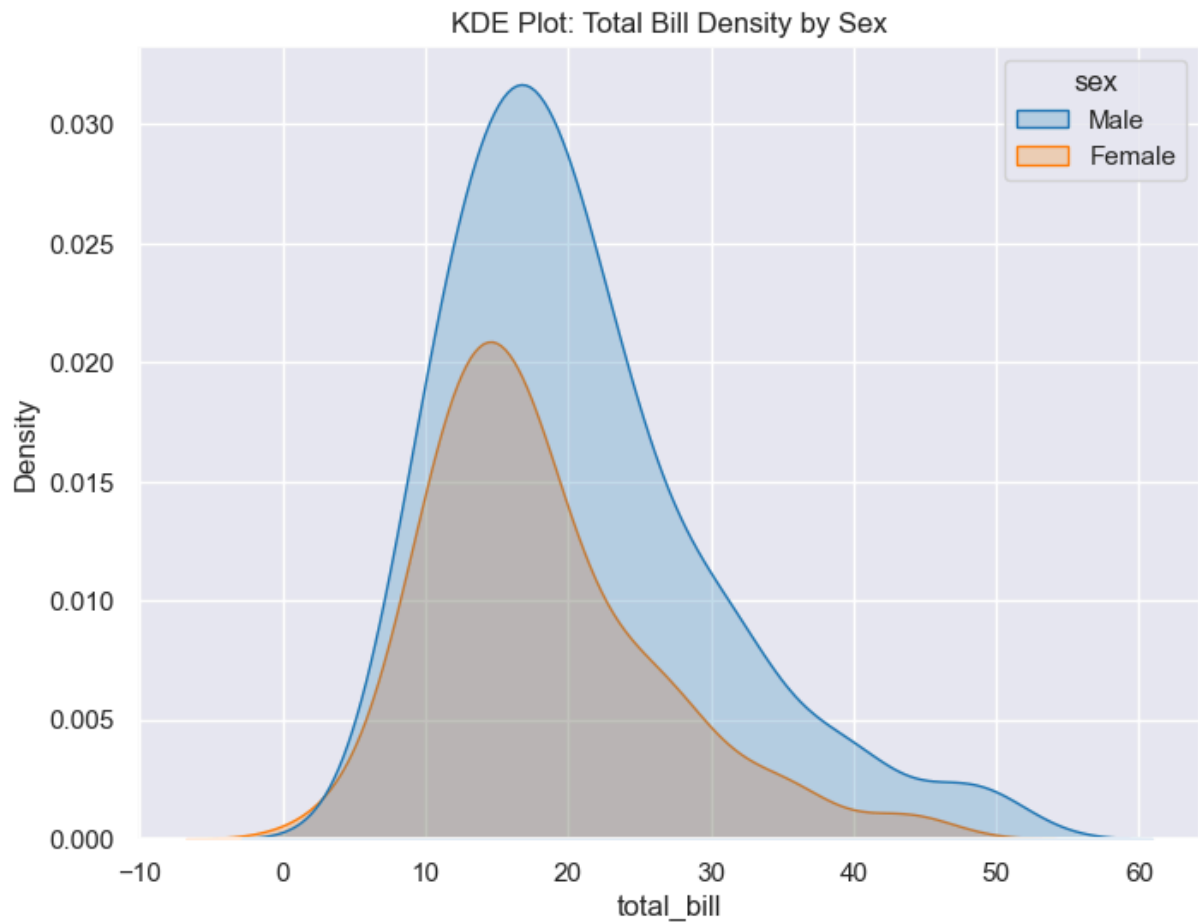
```
In [22]: # 12. FacetGrid: Total Bill by Day, faceted by Time and Smoker
g = sns.FacetGrid(tips, col="time", row="smoker", margin_titles=True)
g.map(sns.histplot, "total_bill", bins=15)
g.fig.suptitle("FacetGrid: Total Bill by Time and Smoker", y=1.02)
plt.show()
```



```
In [23]: # 13. Strip Plot: Tips by Day, colored by Sex
plt.figure(figsize=(8, 6))
sns.stripplot(data=tips, x="day", y="tip", hue="sex", palette="Set1", jitter=True)
plt.title("Strip Plot: Tips by Day and Sex")
plt.show()
```



```
In [24]: # 14. KDE Plot: Total Bill density by Sex
plt.figure(figsize=(8, 6))
sns.kdeplot(data=tips, x="total_bill", hue="sex", fill=True, palette="tab10")
plt.title("KDE Plot: Total Bill Density by Sex")
plt.show()
```



#Explanation of Plots: - Scatter Plot: Shows the relationship between total_bill and tip, with time and size as additional dimensions. - Line Plot: Trends of total_bill across size, split by sex. - Bar Plot: Average total_bill by day, split by sex. - Box Plot: Distribution of tip by day, split by smoker. - Violin Plot: Distribution of total_bill by day, split by time. - Count Plot: Counts of orders by day, split by smoker. - Regression Plot: Linear fit between total_bill and tip. - Histogram: Distribution of total_bill with a KDE curve. - Pair Plot: Pairwise relationships between total_bill, tip, and size, colored by sex. - Catplot (Point): Point estimates of tip by day and sex. - Joint Plot: total_bill vs tip with marginal histograms, colored by smoker. - FacetGrid: total_bill histograms faceted by time and smoker. - Strip Plot: Individual tip points by day, colored by sex. - KDE Plot: Density of total_bill split by sex.

In []:

In []:

In []: